

# Eenheid 1.1

Note Title

2015/01/26

\* Lees Studiegids bl 6

Stewart, Appendix A<sub>2</sub> - A<sub>6</sub>

- Gaan in klas fokus op  
Ongeleikhede

Die res sal jy self moet  
deurgaan.

- Om Wiskunde reg neer  
te skryf is belangrik!!

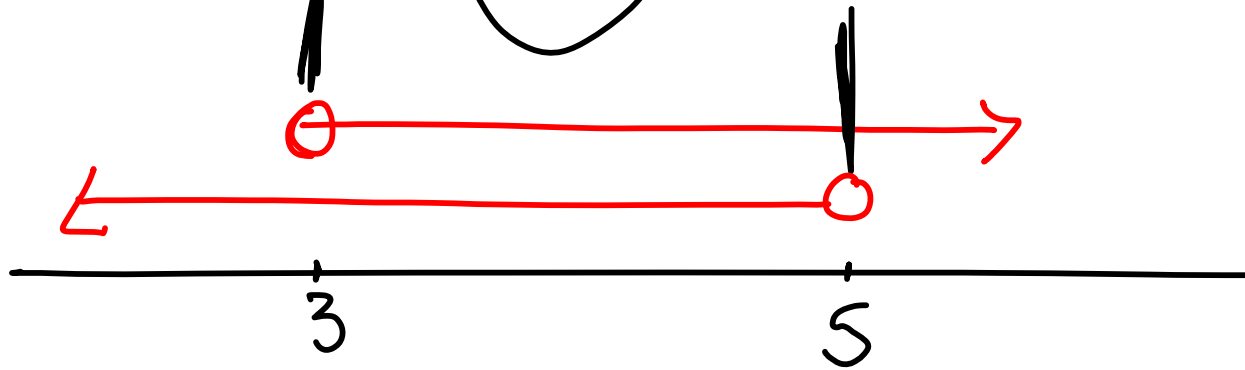
• Notasie en simbole moet  
reg gebruik word.

Vandag kyk ons na:

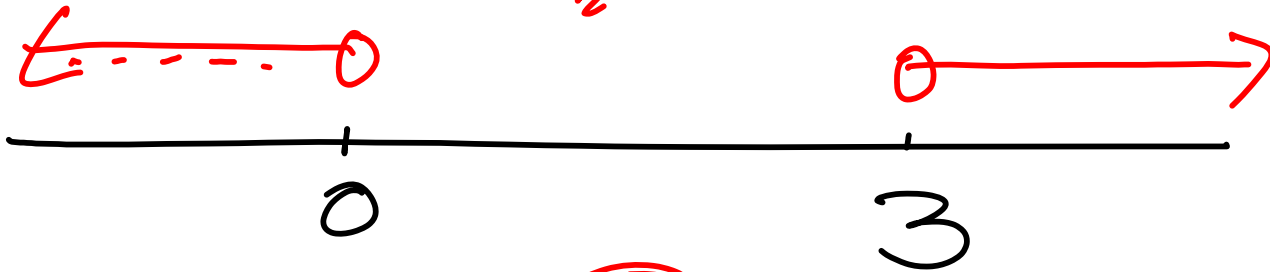
Een simbool; twee woorde  
"  $\Rightarrow$  "  
impliseer en / of

• Impliseer:  $x(x+1) \neq 0 \Rightarrow x \neq 0$  of  $x+1 \neq 0$   
 $\Rightarrow x \neq 0$  of  $x \neq -1$

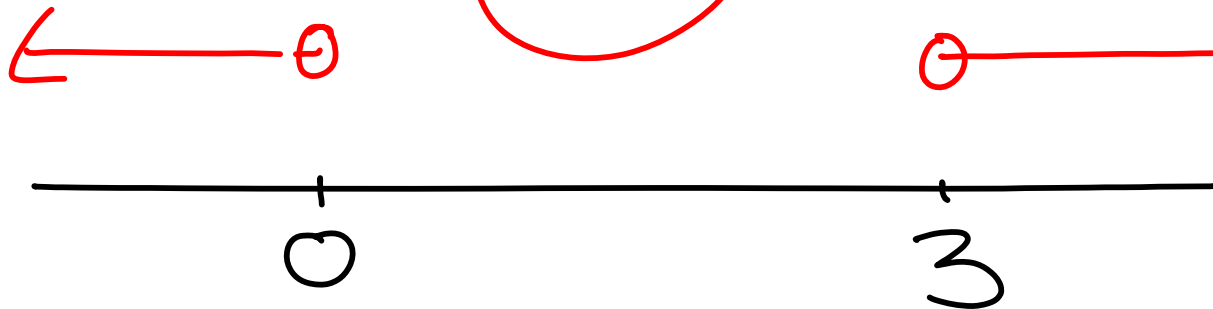
•  $x > 3$  en  $x < 5 \Rightarrow 3 < x < 5$   
 $\Rightarrow x \in (3, 5)$



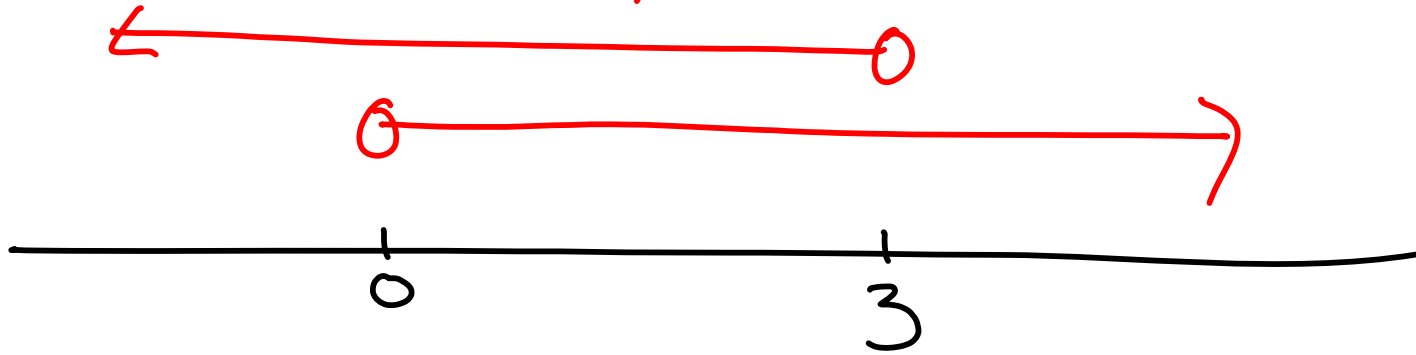
•  $x > 3$  or  $x < 0 \Rightarrow x \in (-\infty, 0) \cup (3, \infty)$



•  $x > 3$  en  $x < 0 \Rightarrow x \in \{\}$   
 $\Rightarrow x \in \emptyset$



•  $x > 0$  of  $x < 3 \Rightarrow x \in \mathbb{R}$



Voorbeelde:

Los op vir  $x$  en skryf jou  
antwoord in intervalnotasie

Onthou as ons praat van  $x$   
bedoel ons  $x \in \mathbb{R}$ .

①

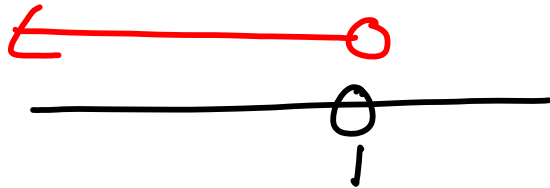
$$\frac{-2}{1-x} < 0$$

↓  
⊕<sub>m</sub>

$$\Rightarrow 1-x > 0$$

$$\Rightarrow -x > -1$$

$$\Rightarrow x < 1$$



$$\Rightarrow x \in (-\infty, 1)$$

- tekens  $< ? \Leftarrow$
- beperkingen ( $x \neq 1$ ) ✓

$$\textcircled{2} \quad \frac{x+1}{2-x} \geq 0^+$$

• ant  $\geq 0$   
 $\downarrow$   
 $= 0$

•  $x \neq 2$

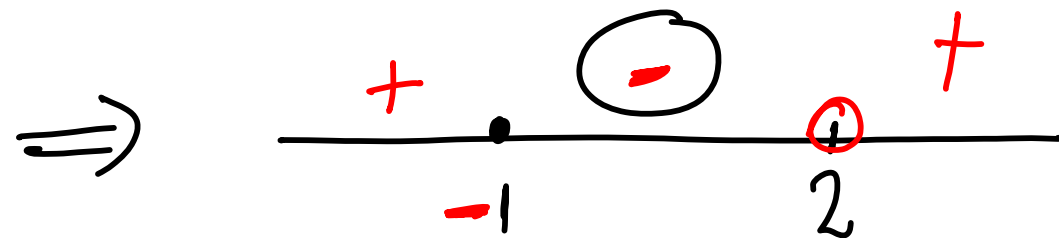
$$\Rightarrow \left( \frac{x+1}{2-x} \right) \times (2-x)^2 \geq 0$$

$$\Rightarrow (x+1) \times (2-x) \geq 0$$

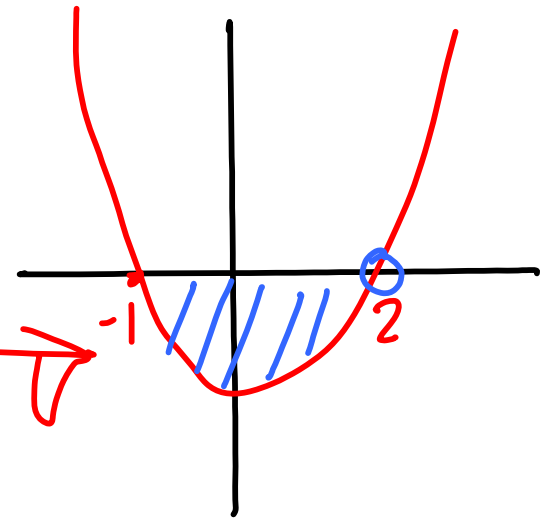
$$\Rightarrow \cancel{2x} - \cancel{x^2} + 2 - \cancel{x} \geq 0$$

$$\Rightarrow x^2 - x - 2 \leq 0$$

$$\Rightarrow (x-2)(x+1) \leq 0$$



$$\Rightarrow x \in [-1, 2)$$

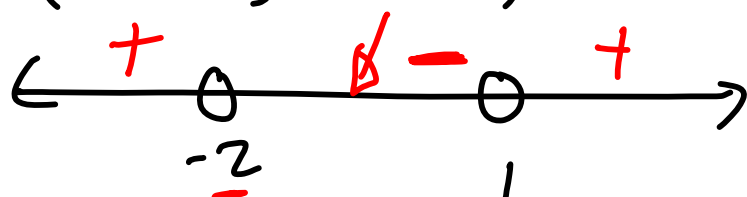


# Kwadratische ongelijkheden

③  $x^2 + x > 2$

$$\Rightarrow x^2 + x - 2 > 0$$

$$\Rightarrow (x+2)(x-1) > 0$$



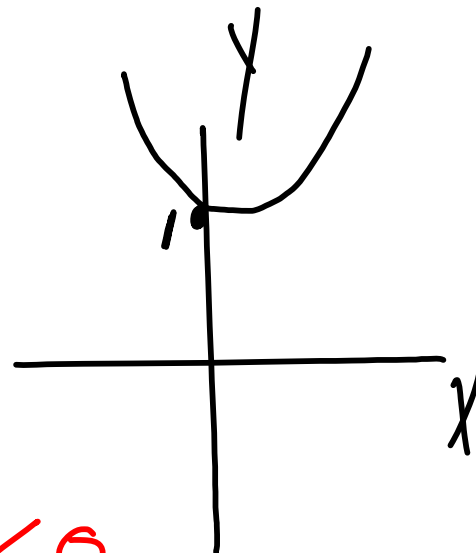
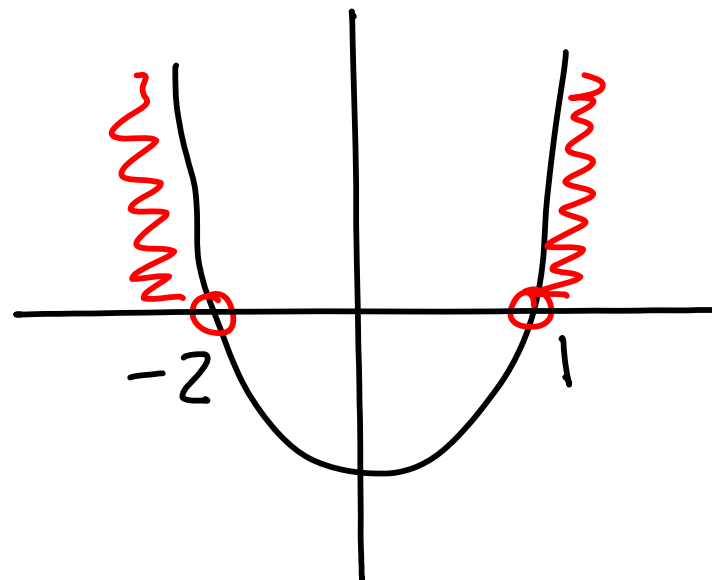
$$\Rightarrow x \in (-\infty, -2) \cup (1, \infty)$$

④  $-x^2 - x - 1 < 0$

$$\Rightarrow x^2 + x + 1 > 0$$

$$\Rightarrow x \in \mathbb{R}$$

$$\Rightarrow x \in (-\infty, \infty)$$



$\Delta = -3 < 0$   
Geen wortels

Wat as:  $x^2 + x + 1 < 0$ ?

$$\Rightarrow x \in \emptyset = \{ \}$$

Onthou geen Formuleblad in  
WTW 158.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Gebruik dit as  
jy nie kan  
faktoriseer nie.

$$\Delta = b^2 - 4ac$$

$$\textcircled{5} \quad x^2 + x - 3 > 0$$

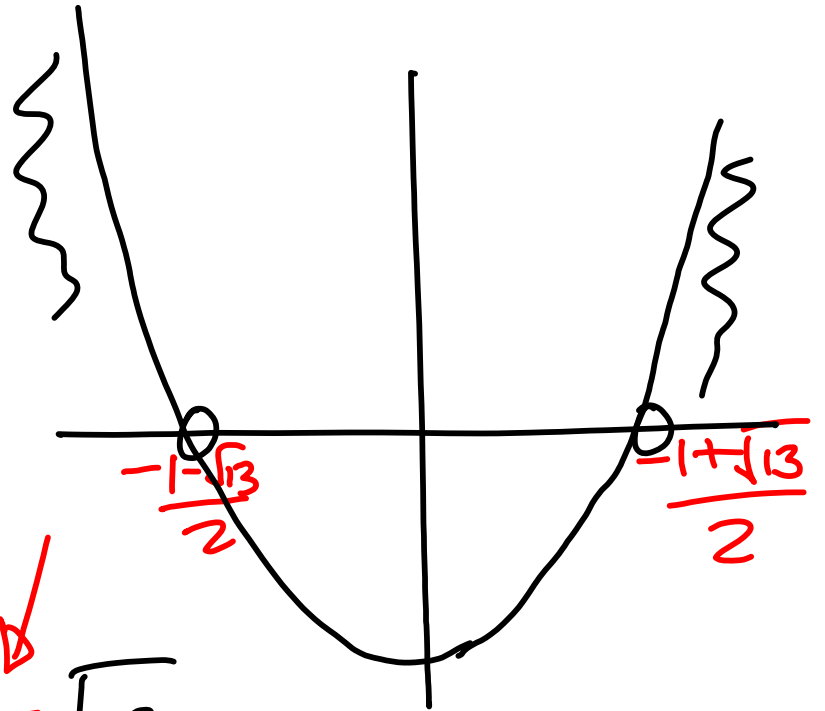
$$x = \frac{-1 \pm \sqrt{1 - (4)(1)(-3)}}{2(1)}$$

$$x = \frac{-1 \pm \sqrt{13}}{2}$$

$$x > \frac{-1 + \sqrt{13}}{2} \quad \text{or} \quad x < -1 - \frac{\sqrt{13}}{2}$$

$$\Rightarrow x \in \left( \frac{-1 + \sqrt{13}}{2}, \infty \right) \cup \left( -\infty, -1 - \frac{\sqrt{13}}{2} \right)$$

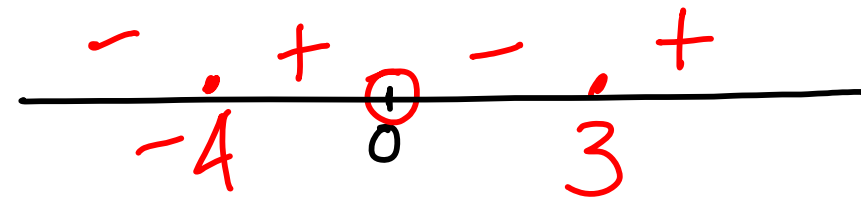
$$\Delta = b^2 - 4ac = \textcircled{13} > 0$$



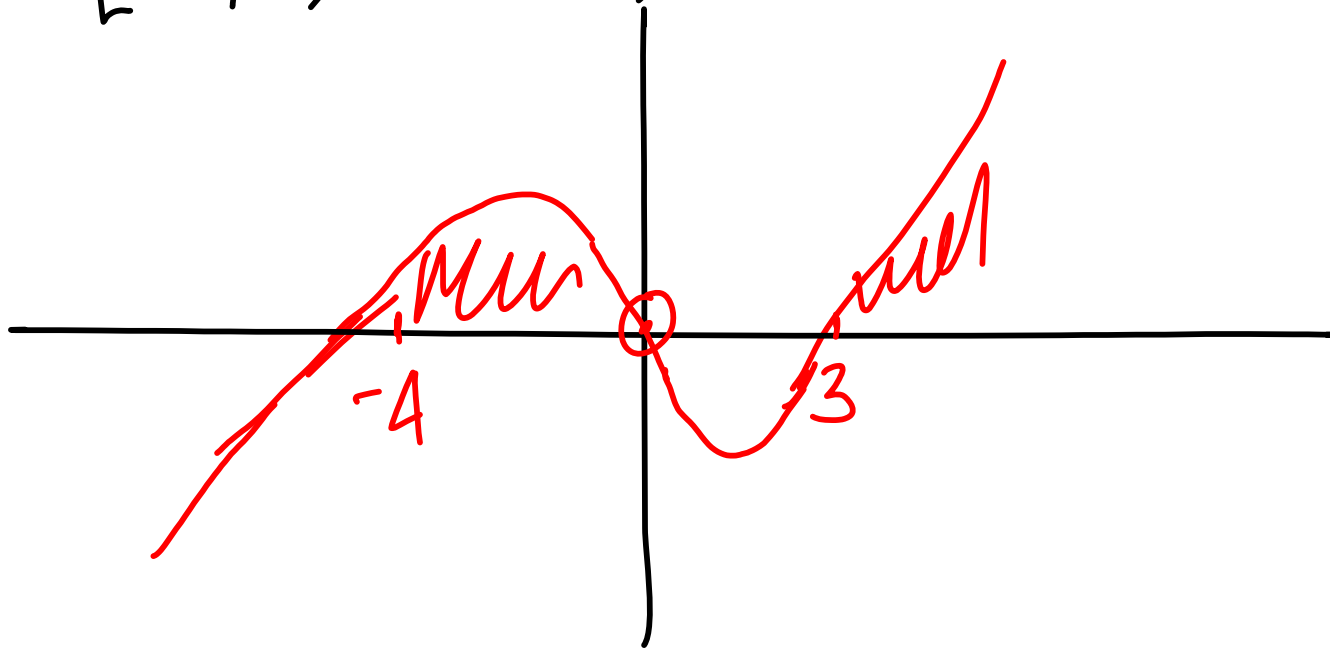
⑥  $\left( \frac{x^2 + x - 12}{x} \right) x^2 \geq 0 \quad x \neq 0$  Onthou:  $x \neq 0$

$\Rightarrow (x^2 + x - 12)x \geq 0$

$\Rightarrow x(x+4)(x-3) \geq 0$



$\Rightarrow x \in [-4, 0) \cup [3, \infty)$



⑦ Stewart A9 #38

$$-3 < \frac{1}{x} \leq 1$$

$$\Rightarrow -3 < \frac{1}{x} \text{ en } \frac{1}{x} \leq 1$$

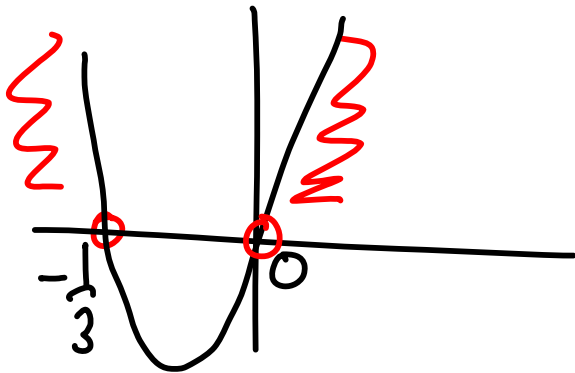
$$\Rightarrow -3x^2 < x \text{ en } x \leq x^2$$

$$\Rightarrow 3x^2 + x > 0 \text{ en } x^2 - x \geq 0$$

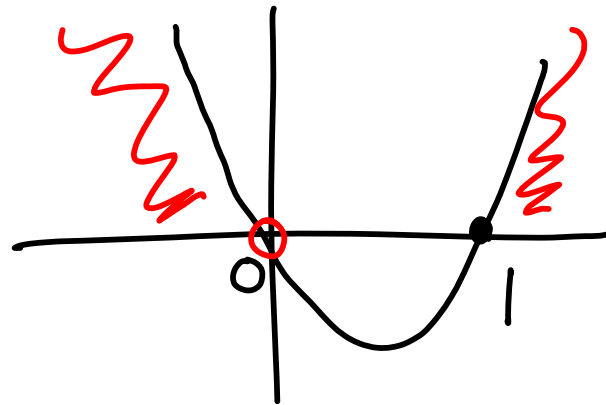
$$\Rightarrow x(3x+1) > 0 \text{ en } x(x-1) \geq 0$$

En orthou?  
 $x \neq 0$

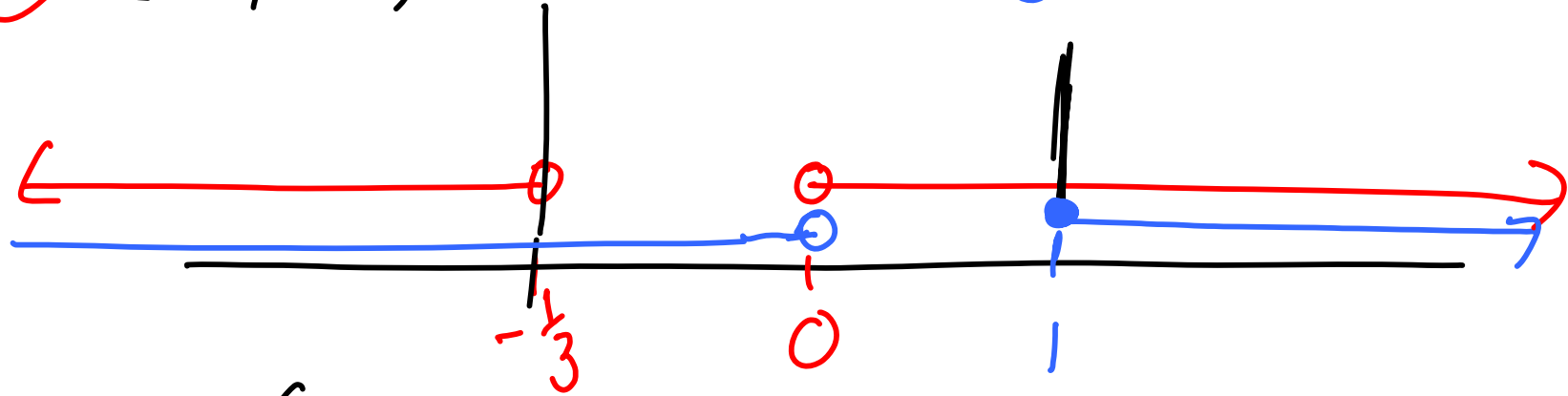
(en  $x \neq 0$ )



en  
 $\approx$



$$\Rightarrow \textcircled{x \in (-\infty, -\frac{1}{3}) \cup (0, \infty)} \text{ or } \textcircled{x \in (-\infty, 0) \cup [1, \infty)}$$



$$\Rightarrow x \in (-\infty, -\frac{1}{3}) \cup [1, \infty)$$